Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Scientist & Engineer with 3 years of industry experience building reliable data pipelines, schemas and analytical products in Python and SQL, and an MSc in Data Science from the University of Bristol (awaiting result). Ingested the **Morningstar API** into a **Normalized PostgreSQL schema** integrated with legacy data, built a **fuzzy-matching** email pipeline that saves **60 analyst hours a month**, and co-built the data preprocessing and **calibrated scoring and ranking algorithm** for an MSc dissertation with AstraZeneca (14 clean tables from 17 inputs, identity resolution, integrity checks). Built **embeddings** and **RAG** over internal documents with department-based access, **role-based access control** and Compliance review of every release, analysed data with **Pandas** and **Market Basket Analysis**, and worked directly with Client Service, Compliance, Investment and Marketing teams; twice recognised internally with Best Team Project (2024, 2025). Seeking to bring this to Norton Rose Fulbright's R&D team to build governed, reusable data assets and models on Microsoft Fabric and Databricks.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Pipelines, Schemas & Data Quality

- Designed a **Normalized SQL Schema** (**PostgreSQL**) for the internal analytics platform and wrote **Python** scripts ingesting the enterprise-licensed **Morningstar API** for every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity) and integrated with the company's legacy Excel data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures** for preprocessing and data insertion.
- Built a scheduled **Python** pipeline (**EventBridge**, **Lambda**, **AWS SageMaker**) that turns over **300** daily credit rating agency emails into structured data, **fuzzy-matches** company names to the analysts' tracked list and drops the rest, and sends a consolidated daily summary of previous and changed ratings with an Excel copy in **S3**, **saving 60 manual analyst hours a month**; upgraded parsing from **AWS Textract** to **Azure OpenAI** when a rating agency began sending emails as images.
- Reused the curated data across products: GenAI agents call **Python** functions that run **SQL** queries over it for fund comparison.

Analysis, Modelling & NLP

- Analyzed customer data using **Python Pandas** and identified customer buying patterns with **Market Basket Analysis** (**Apriori Algorithm**), handing the insights to the marketing team, who decided how to act on them.
- Designed and built **GenAI** applications end to end (**ReactJS**, **Python FastAPI**, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) to answer department-specific queries over internal documents, with an **embeddings** pipeline into a separate **pgvector** database per department; first as the only engineer alongside my manager, later leading the team.
- Developed **Power BI** dashboards for fortnightly credit report analysis for the investment team, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**; recreated them in **ReactJS** for wider access at lower cost than scaling Power BI licences.

Governance, Deployment & Collaboration

- Handled sensitive data responsibly: department-based document access, **role-based access control** with a **Python** authentication function against the company **LDAP** server, **AWS Security Groups** limiting access to company private IPs, Compliance review of every release and prompts versioned in **Git**; chose models on cost and fit, with **self-hosted Llama-3** for the largest department and stronger **Azure OpenAI** outputs for decision makers.
- Built one-click deployment to dev and prod using **Jenkins** and **GitLab** in a **CI/CD** pipeline, with **Docker** microservices on **AWS** (**EC2**, **S3**, **ECR**, **Lambda**) and separate Git repositories and dev/prod branches for frontend and backend.
- Worked directly with business stakeholders in Client Service, Compliance, Investment and Marketing to turn ambiguous requirements into working tools, released them to real users and improved them on their feedback, and explained results to non-technical audiences; acted as **de facto team lead** for up to 4 interns in my final year; recognised internally with **Best Team Project** (2024, 2025).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the data preprocessing pipeline, the scoring algorithm and the RAG-based AI assistant for a tool that ranks cancer cell lines by target inclusion and exclusion genes. The pipeline builds 14 clean, joined tables from 17 raw and augmented inputs (DepMap, GEO, HPA), with identity resolution across sources, missingness handling, integrity checks and an audit manifest; per-gene scoring across six evidence layers, weighted by correlation and calibrated across lineages, shows the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Multi-phase architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating natural-language image descriptions, evaluated with SBERT cosine, BLEU and ROUGE-L across five seeds and benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Swindon Borough Council Fuel-Poverty Hack:** household and neighbourhood-level fuel poverty data analysis tool built for the council's hackathon. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git

SKILLS

- **Tools:** Power BI, Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker, Textract), Git, GitLab, Jenkins, Docker, Azure OpenAI, LlamaIndex, pgvector, Claude Code, Nginx, Linux, VS Code.
- **Languages:** Python (pandas, scikit-learn, fuzzywuzzy, FastAPI), R (dplyr, tidyverse, ggplot2), SQL, NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector), SQLite